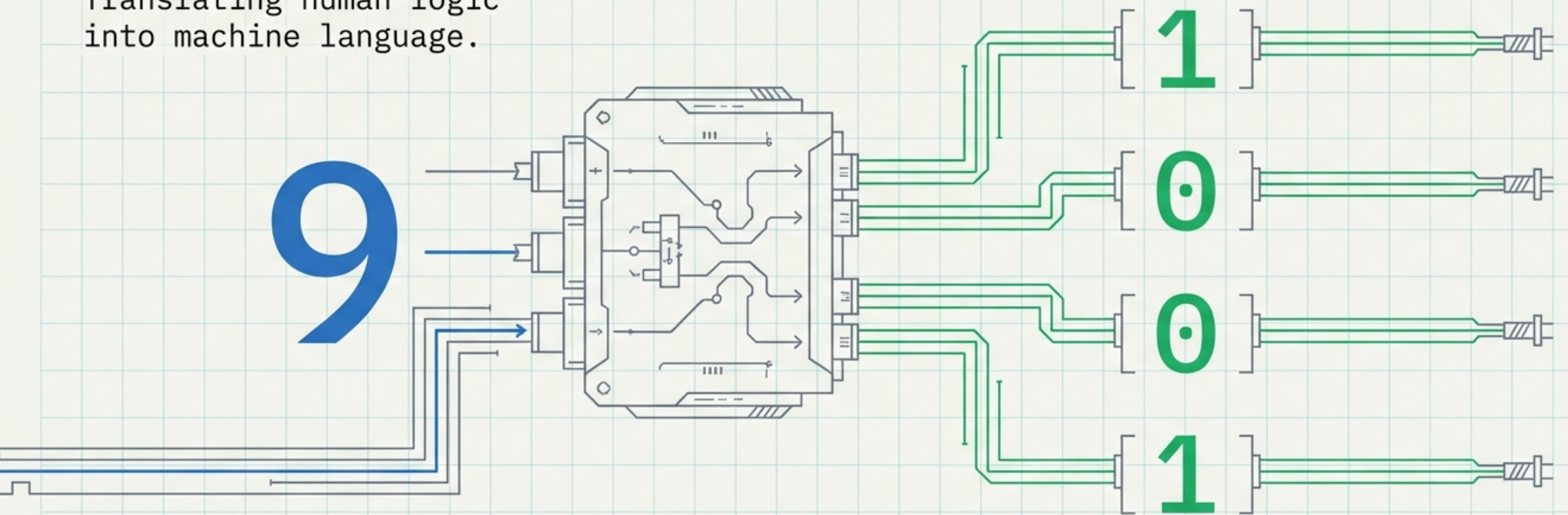


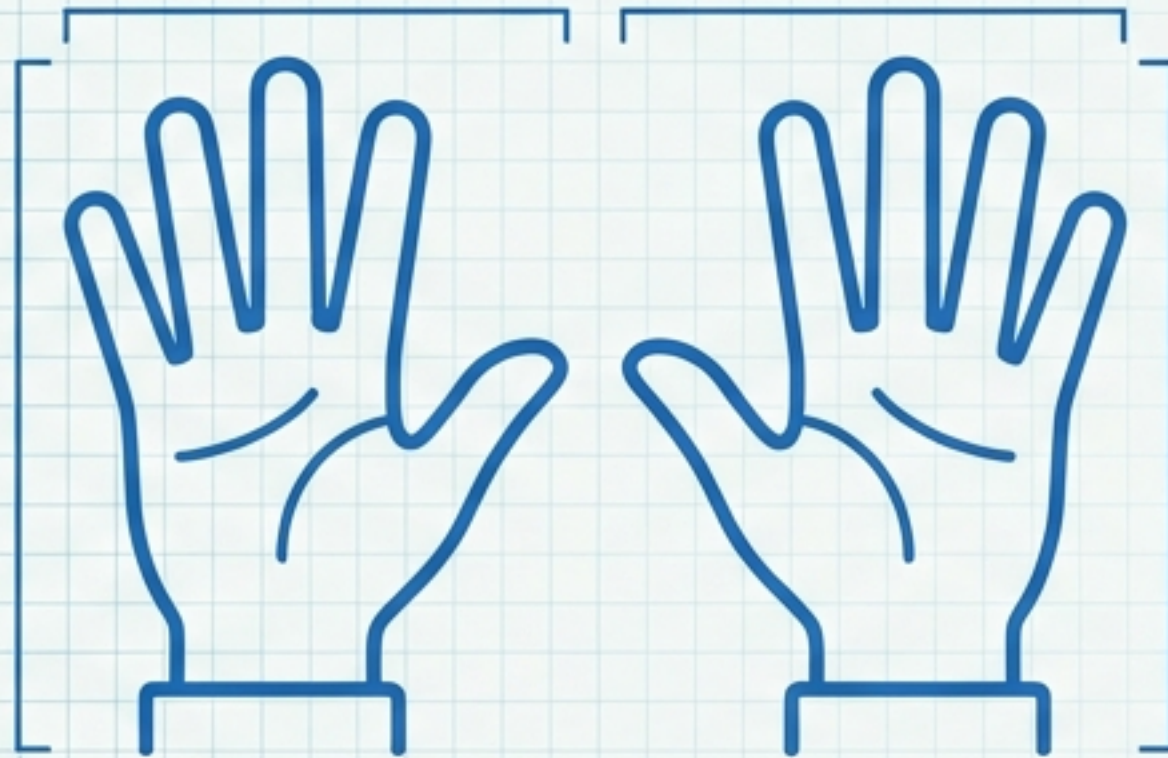
The Architecture of Number Systems

Translating human logic into machine language.



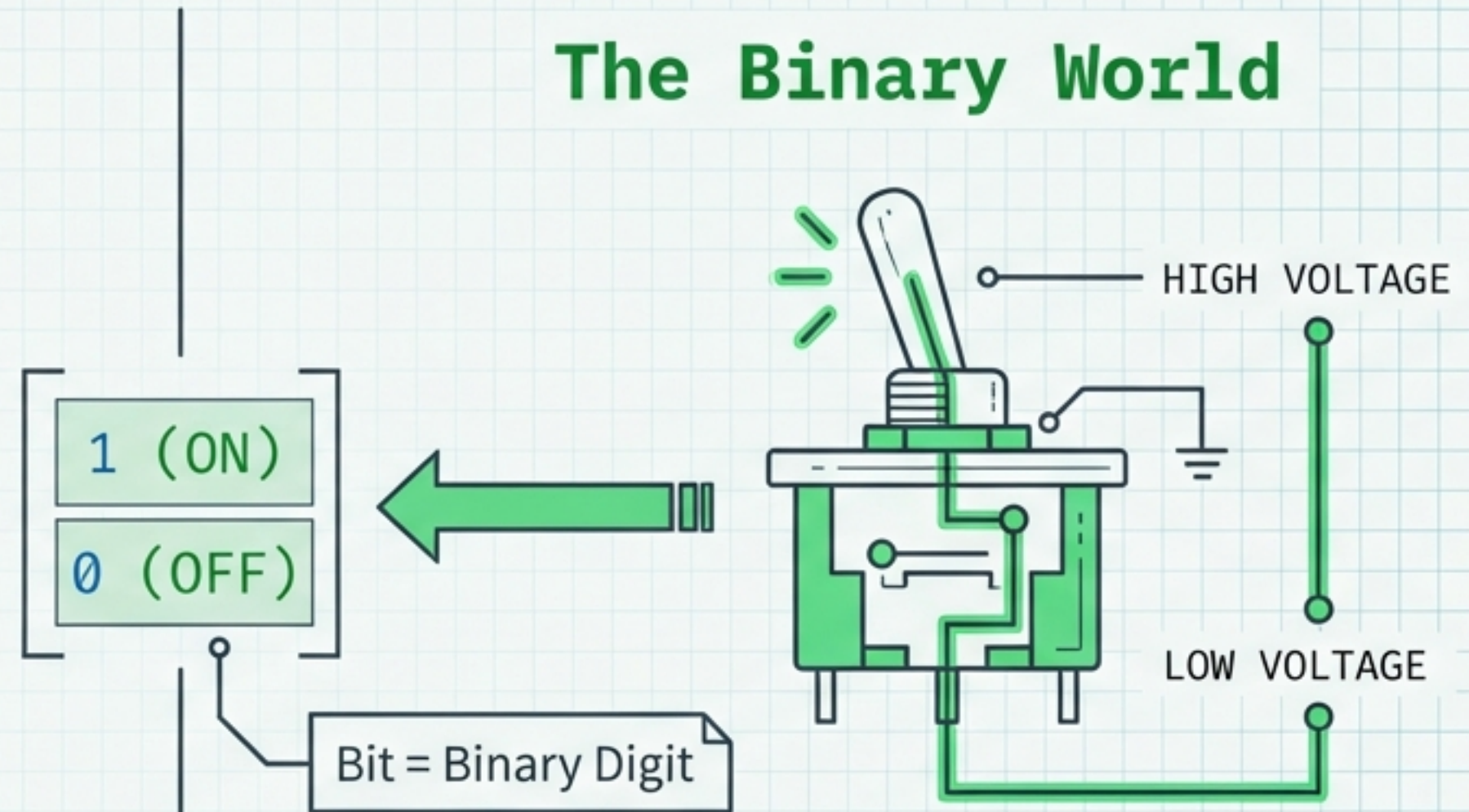
The Language Barrier of Hardware

The Decimal World



Humans process the world in **Base-10** (Decimal) using digits **0-9**. This is an abstract mathematical concept tailored to human biology.

The Binary World



Computers are electrical systems built on digital circuits. They possess only two physical states: High Voltage (ON) and Low Voltage (OFF).

The Base Matrix

DECIMAL	BINARY	OCTAL	HEXADECIMAL
10	2	8	16
0 to 9	0, 1	0 to 7	0 to 9, A to F
Human input/output and daily life.	Core machine language, data processing, & hardware storage.	Legacy computing; early binary data grouping.	Representing memory addresses and shrinking long binary strings.
523	1010011	157	0x2A

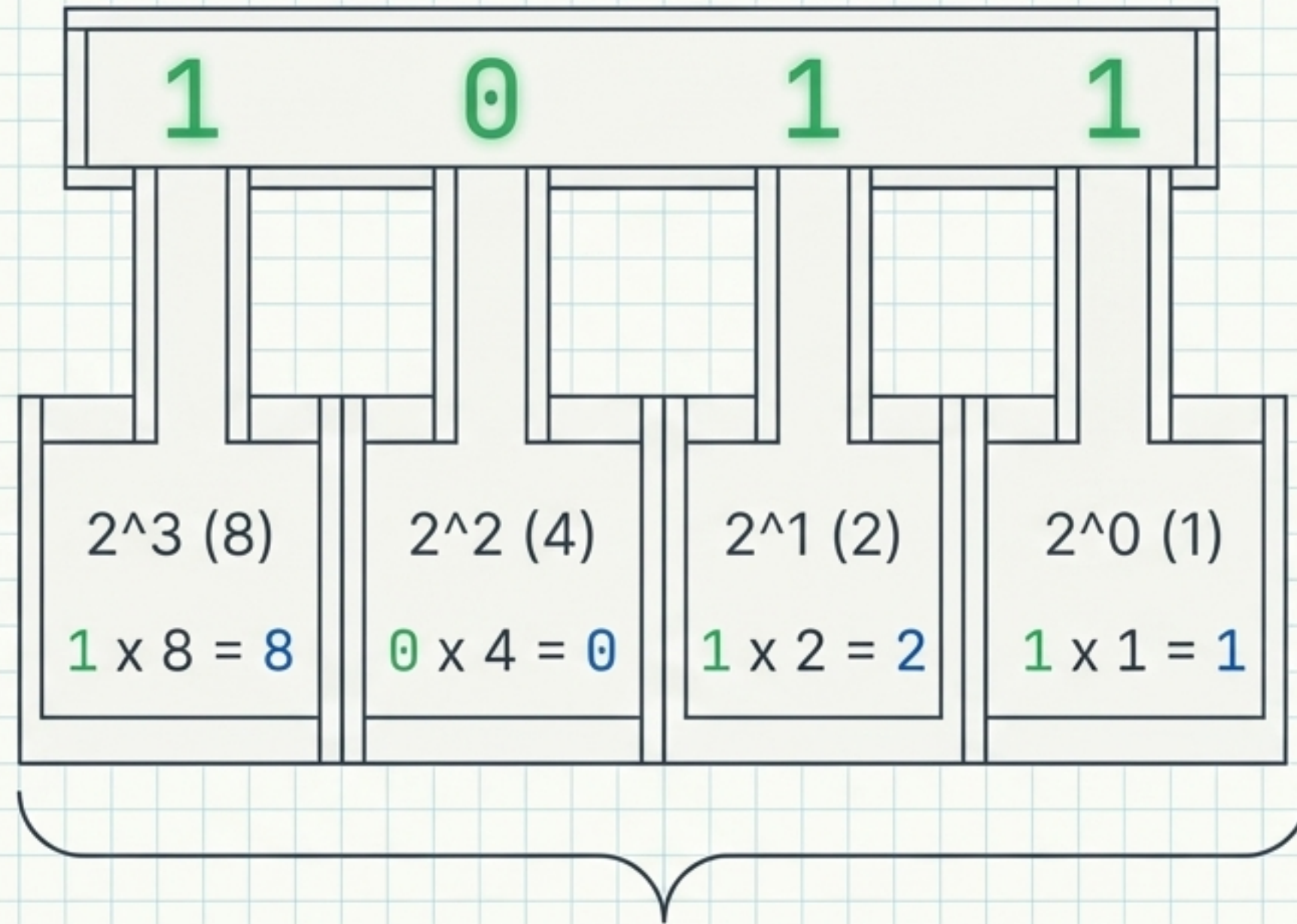
The Universal Rule: Positional Weight

Before converting bases, we must understand how all number systems are constructed. Every digit's value is multiplied by its positional power.

$$\begin{array}{ccc} 5 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 5 \times 10^2 & 2 \times 10^1 & 3 \times 10^0 \\ \downarrow & \downarrow & \downarrow \\ 5 \times 100 & 2 \times 10 & 3 \times 1 \\ \downarrow & \downarrow & \downarrow \\ 500 & 20 & 3 \\ 500 + 20 + 3 = 523 \end{array}$$

This identical logic applies to every base system. Only the base number (10) changes.

Decoding Binary to Decimal



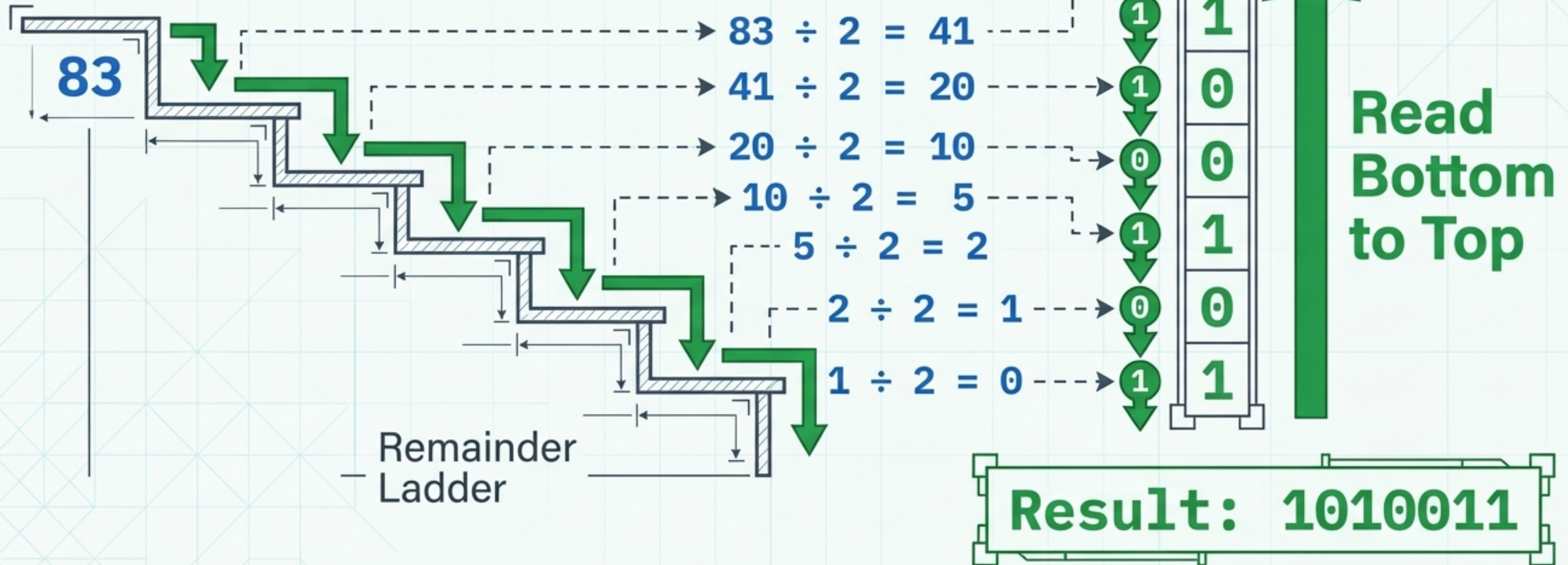
$$8 + 0 + 2 + 1 \rightarrow$$

Result Plate

$$= 11$$

Encoding Decimal to Binary

To convert from human logic (Base-10) to machine logic (Base-2), **divide by 2 repeatedly** and **collect the remainders**.



The Constraints of Base-8 (Octal)

Octal operates entirely within digits 0 through 7.

Validation Error

~~198~~₈



Logical Error: The digits 8 and 9 do not exist in the Octal system. If you see an 8 or 9, it cannot be Base-8.

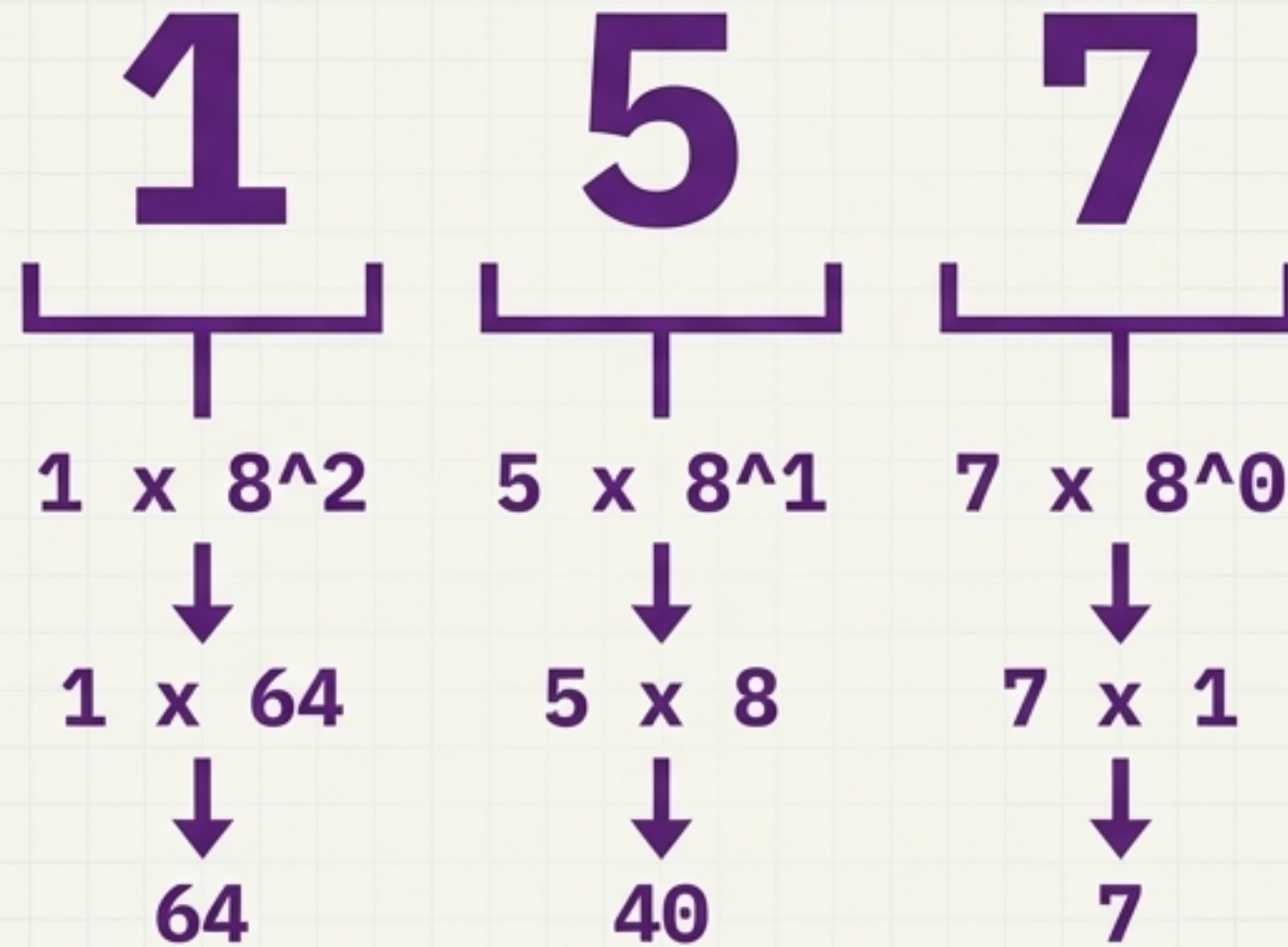
Valid Example

157₈



Octal to Decimal: The Universal Rule Applies

Anatomy of an Octal Number



$$64 + 40 + 7 \rightarrow$$

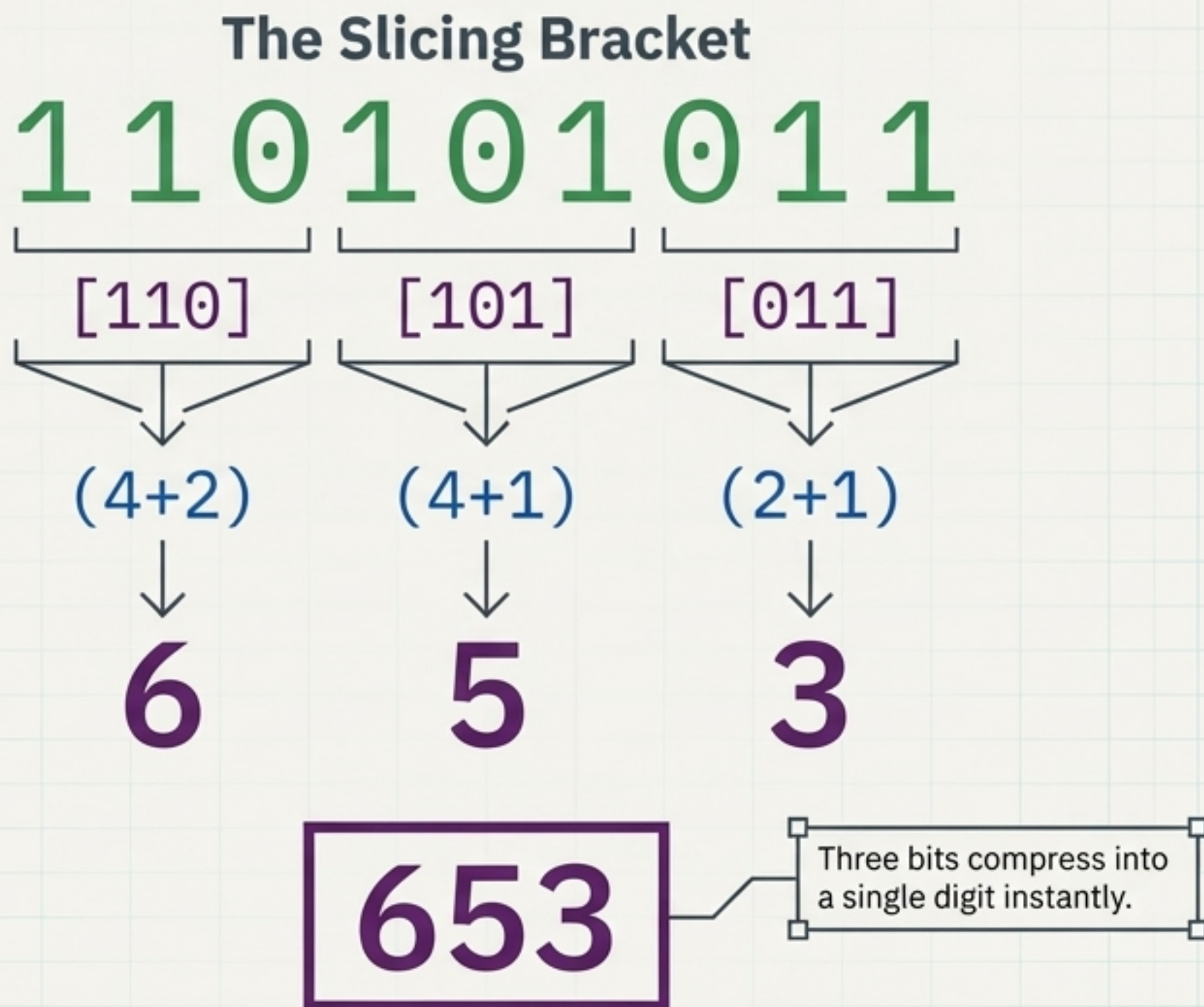
Result Plate

$$= 111$$

Instead of powers of 2 or 10, we simply use powers of 8.

The 3-Bit Bracket (Binary to Octal Shortcut)

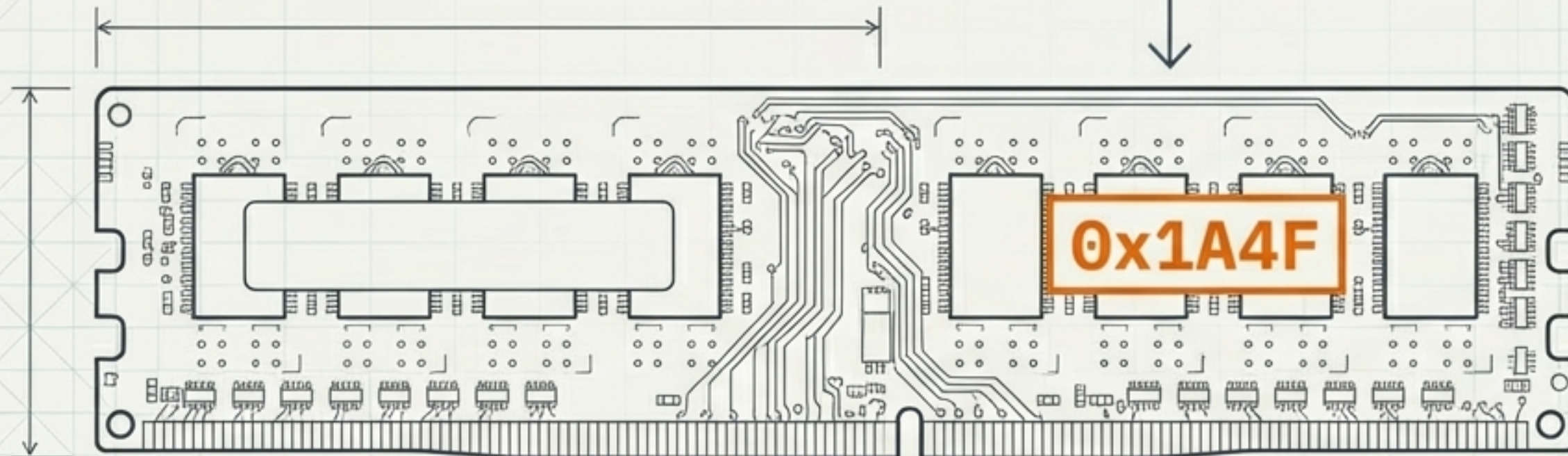
Why does Octal exist? Because $2^3 = 8$. Exactly three binary bits map perfectly to one octal digit, making it an ideal shortcut for long strings of code.



Hexadecimal (Base-16) & The Alphabet

When representing hardware memory addresses, even Octal is too long. We need Base-16. But humanity only invented 10 digits (0-9). The solution? Borrow the alphabet.

	A (10)	B (11)	C (12)	D (13)	E (14)	F (15)
0, 1, 2, 3, 4, 5, 6, 7, 8, 9	A	B	C	D	E	F

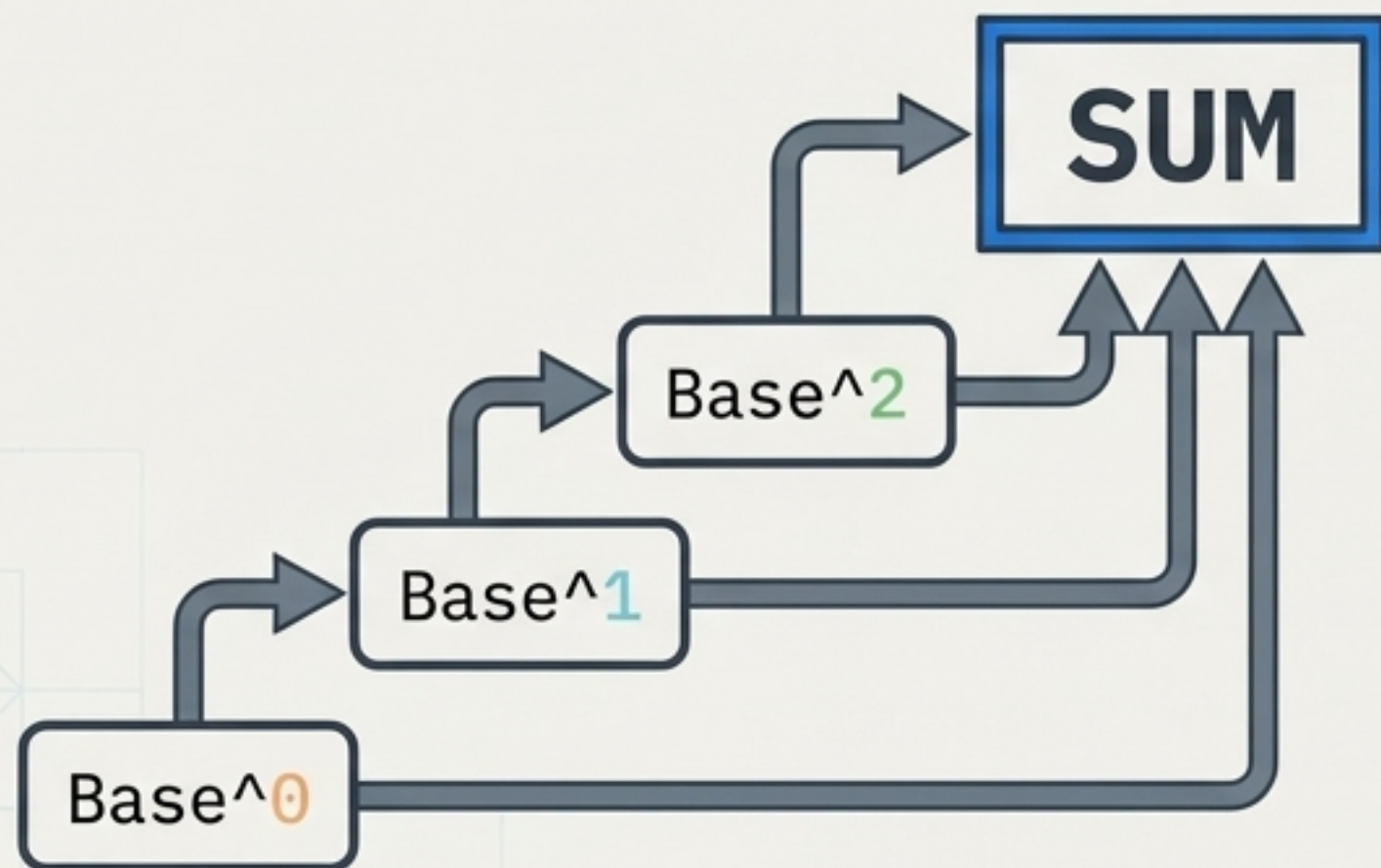


Real-world Memory Allocation

The Two Core Algorithms

Moving TO Decimal

(e.g., Binary/Octal -> Decimal)

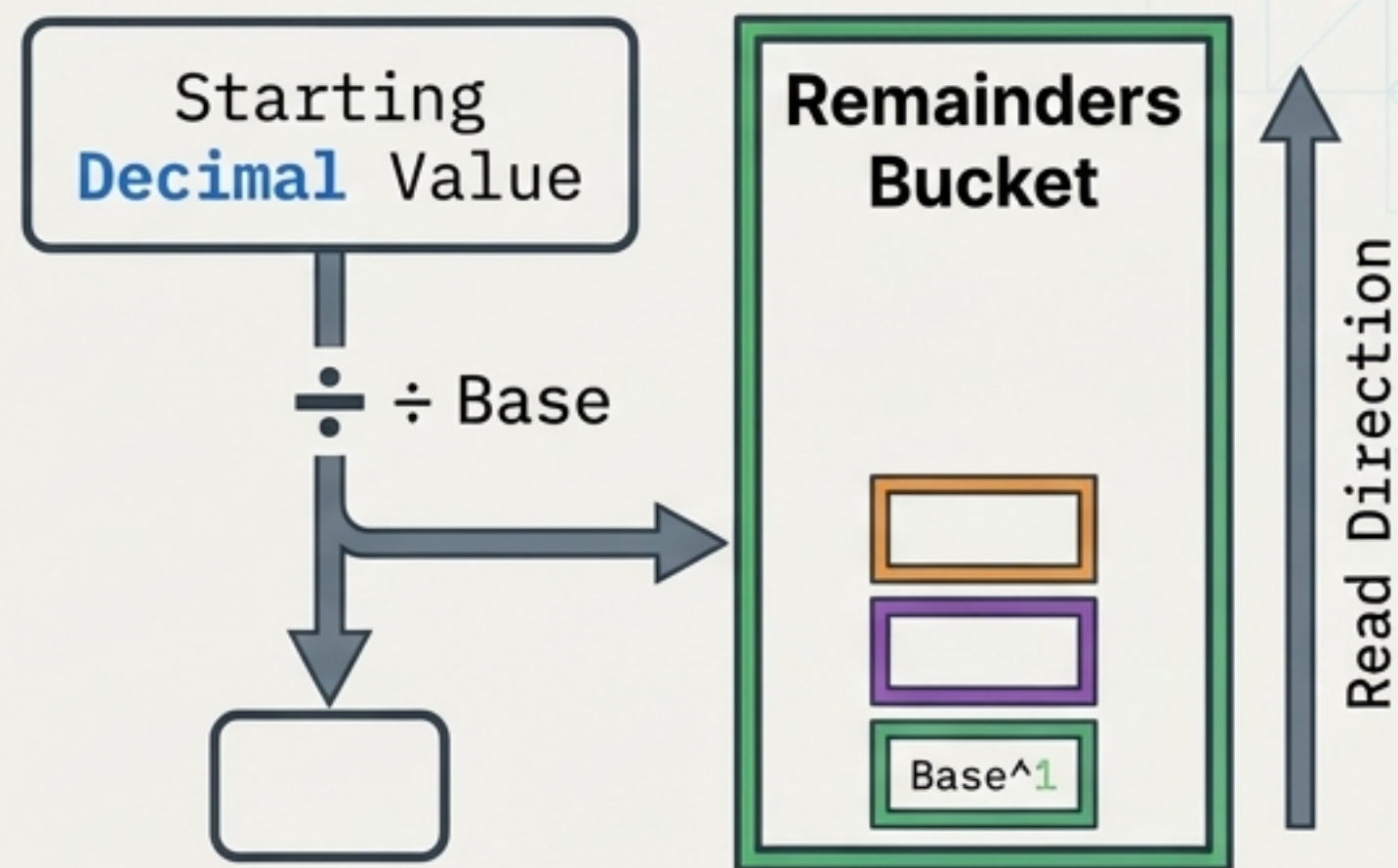


Key Rule:

Multiply and Add.

Moving FROM Decimal

(e.g., Decimal -> Binary)



Key Rule:

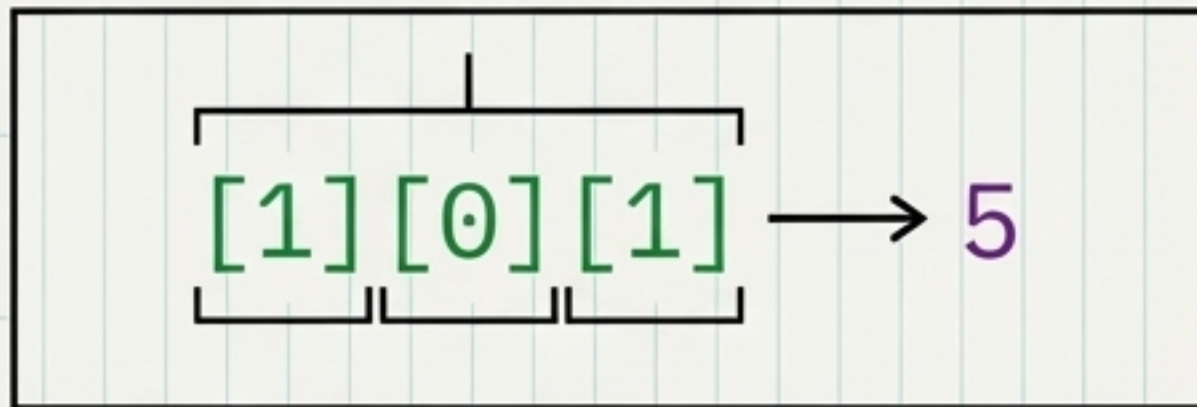
Divide and Collect Remainders (Bottom-to-Top).

The Blueprint Cheat Sheet

The Bases Table

Base	Digits	Power Sequence
Binary (Base-2)	0, 1	1, 2, 4, 8, 16...
Octal (Base-8)	0-7	1, 8, 64...
Decimal (Base-10)	0-9	1, 10, 100...
Hexadecimal (Base-16)	0-9, A-F	1, 16, 256...

The Shortcut Visual



3 binary bits = 1 octal digit

The Golden Rule

The position of a digit matters just as much as the digit itself.
Hardware only speaks Binary.
Everything else is human convenience.